[INCH-POUND]

MIL-DTL-11732E
25 January 1999
SUPERSEDING
MIL-C-11732D
20 March 1973
MIL-C-11732/1
8 March 1973

DETAIL SPECIFICATION

CYLINDER, COMPRESSED GAS: CHLORINE; DOT 3A480 OR DOT 3AA480, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers four types of steel cylinders, with valve and cap, intended for chlorine service.

1.2 Classification. The cylinders covered by this specification will be of the following types, as specified (see 6.2).

Type I - 20 pounds

Type II - 30 pounds

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610 by using Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8120

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Type III - 100 pounds

Type IV - 150 pounds

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

BB-N-411	-	Nitrogen, Technical
TT-C-490	-	Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings
TT-E-489	-	Enamel, Alkyd, Gloss, Low VOC Content
TT-P-664	-	Primer Coating, Alkyd, Corrosion-Inhibiting, Lead and Chromate Free, VOC Compliant

DEPARTMENT OF DEFENSE

MIL-DTL-2	-	Valves, Cylinder, Gas (for Compressed or Liquefied Gases), General Specification for
MIL-DTL-2/20	-	Valve, Cylinder, Gas: Chlorine, Outlet 820
MIL-C-17376	-	Caps and Flanges, Compressed-Gas Cylinder, General Specification for

STANDARDS

FEDERAL

FED-STD-H28	-	Screw-Thread Standards for Federal Services

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DEPARTMENT OF DEFENSE

MIL-STD-101 - Color Code for Pipelines and for Compressed Gas Cylinders

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents and drawings. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issue is that cited in the solicitation.

DEPARTMENT OF TRANSPORTATION

Code of Federal Regulations, Title 49-Transportation (49 CFR), Parts 100 to 185

(Application for copies of the above document are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0002.)

DRAWING

810-1385867 - Decalcomania for Navy Gas Cylinders

(Copies of drawings required by suppliers in connection with a specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

STEEL STRUCTURES PAINTING COUNCIL (SSPC)

SSPC SP5 - White Metal Blast Cleaning (DoD adopted)
SSPC SP6 - Commercial Blast Cleaning (DoD adopted)
SSPC SP10 - Near White Blast Cleaning (DoD adopted)

(Application for copies of SSPC documents should be addressed to the Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburgh, PA 15213.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Configuration. Cylinders shall conform to requirements for 49 CFR specification 3A or 3AA, and shall be as specified herein.

3.1.1 Cylinders. The cylinder shall be as specified in table I. All dimensions are in inches. The tolerance for “D” is plus or minus 1/2 inch and that for “H” is plus or minus 2 inches.

TABLE I. Cylinder configuration.

Type	Nominal gas capacity (pounds)	Outside dimensions D (inches)	H (inches)	Min. water capacity (pounds)	Max. tare weight (pounds)
I	20	6	19	16	25
II	30	6	29	24	35
III	100	8-1/2	48	80	85
IV	150	10-1/4	48	120	120

3.1.2 Flanges, valves, and caps. The cylinder shall be supplied with flanges, valves and cap in accordance with figure 1.

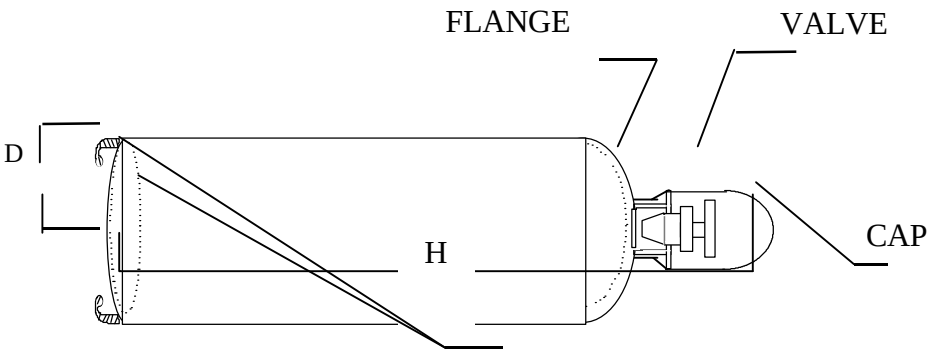

FIGURE 1. Cylinder configuration.

3.1.3 Bottom design. The bottom design shall be humped back or shall be designed with a foot ring for upright stability. Foot rings, when used, shall have a minimum of two ventilation holes and two drainage holes equally spaced.

3.2 Construction. Cylinders shall be true of form and shall be stable when in an upright position. The neck of each cylinder shall be tapped internally 3/4-14 NGT (C1)-1. Threads shall conform to FED-STD-H28.

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3.3 Cylinder processing.

3.3.1 Internal surfaces. The internal surfaces of drawn cylinders, prior to necking, shall be free of mill scale, burrs, oxidation products and all foreign matter. When required, only iron-based abrasives shall be used to remove unstable surfaces or oxidation products from the cylinder wall. Metal discoloration, rust bloom, and fixed scale shall not be considered criteria for cylinder rejection.

3.3.2 Cylinder cleaning. After necking, drilling, threading and deburring, the cylinders shall be cleared of loose particles by vacuum or by inverting the cylinder and use of a compressed air sweep (see 6.3). The cylinder shall be flushed with hot fluids when necessary to remove grease, oil, and fat from processing. Chemicals that will react with chlorine shall not be used.

3.3.3 Cylinder drying. Internal cylinder moisture, from processing or hydrostatic testing, shall be removed by inverting and steaming the surfaces until hot to the touch, followed by drying with filtered air having a dew point less than 30 °F. In lieu of filtered air, drying by suction of ambient air through the cylinder is allowed. The residual air in the dry cylinder shall be replaced by introducing dry nitrogen gas conforming to BB-N-411, grade B, through a bayonet extending to the bottom end of the cylinder. Closure shall be carried out immediately to limit atmospheric interchange.

3.4 Closure. Cylinders shall be equipped with a flange and a cap conforming to MIL-C-17376. A plug or closed valve conforming to MIL-DTL-2/20, as specified, shall be installed in the neck of the cylinder (see 6.2).

3.5 Tare weight. The tare weight of the cylinder shall be the weight of the cylinder, flange, and valve, but exclude the valve protection cap. The tare weight shall be expressed in pounds and quarter-pounds, expressed in 4-ounce increments to the nearest quarter-pound. The figures expressing pounds shall be separated from the figures expressing ounces by a dash (-).

3.6 Test date. The cylinders shall be delivered within 1 year of the test date stamped on the cylinder.

3.7 Cylinder identification. Marking shall be in characters not less than one-fourth inch high.

3.7.1 Standard markings. Standard markings shall be plainly and permanently marked by stamping on the shoulder of each cylinder starting near the neck as follows:

a. “DOT-3A480” or “DOT-3AA480” in characters not less than three-eighths inch high for cylinders more than 6 inches in outside diameter.
b. Serial number (see 6.6), as registered with the Bureau of Explosives, in characters not less than one-half inch high for cylinders not more than 6 inches in outside diameter.
c. “US GOVT,” Government symbol, as registered with the Bureau of Explosives, in characters not less than one-half inch high for cylinders more 6 inches in outside diameter.
d. The inspector’s official mark.

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e. Date of test, stamped diametrically opposite the above marking.
f. Manufacturer's or retester's mark, near the date of test (preferably right above it).
g. The tare weight shall be stamped in a position other than in sequence with the serial number or test date.

3.7.2 Additional markings. In addition to the above markings, the cylinders shall bear the following markings:

a. The name of the gas, stenciled on the cylinder in accordance with MIL-STD-101.
b. When specified (see 6.2), two decalcomanias conforming to drawing 810-1385867 shall be affixed diametrically opposite to one another on the sides of the cylinder, 90 degrees from the stenciled name of the gas.

3.8 Treatment and painting.

3.8.1 Cleaning. Each cylinder and cap shall be cleaned in accordance with TT-C-490. Cleaning may be accomplished by:

a. Chemical methods such as solvent cleaning, alkaline cleaning, acid cleaning, pickling, descaling with hydride, or paint stripping; or,
b. Electrochemical cleaning methods such as electrolyte alkaline, electropolishing, or electrolytic pickling; or,
c. Mechanical means such as blasting, chipping, wire brushing, or grinding.

3.8.1.1 Blasting. Where blasting is appropriate, blast in accordance with Steel Structures Painting Council (SSPC) SSPC-SP6, unless SSPC-SP5 or SSPC-SP10 is specified (see 6.2). Blasting shall not be used on surfaces that could be damaged by blasting or on components containing such surfaces unless protective masking has been applied.

3.8.2 Treatment. Cylinders shall be treated as soon as possible after cleaning, as specified in 3.8.1, with a zinc phosphate chemical conversion coating in accordance with TT-C-490, type I.

3.8.3 Painting. Coatings shall be applied in an ambient temperature of not less than 50 °F nor more than 90 °F. Paint shall be applied by any method that will deposit the specified film coat thickness and comply with any applicable transfer efficiency regulations. Paint and surface shall be approximately the same temperature except when hot spray is used.

3.8.3.1 Primer. Primer conforming to TT-P-664 shall be applied to each dry, clean cylinder as soon as practicable after cleaning and treatment (see 3.8.1 and 3.8.2). Dry film thickness of primer shall be 0.7 to 1.0 one-thousandth of an inch (mil).

3.8.3.2 Finishing coat. Finishing coat conforming to TT-E-489 shall be applied to each cylinder, color conforming to the cylinder color code requirement of MIL-STD-101. Dry film thickness of finishing coat shall be 1.0 to 1.5 mil.

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3.9 Workmanship. Cylinder, valve, plug, flange, and cap shall be clean and free from grit and loose scale. Edges shall be rounded or chamfered. Cylinder shall be cleaned and free of other surface defects detrimental to the intended use.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as conformance inspection.

4.2 Conformance inspection.

4.2.1 Examination. Each cylinder shall be examined for defects as listed in table II.

TABLE II. Processing defects.

Defect	Requirement
Cylinder not in accordance with 49 CFR specification for a DOT3A or DOT3AA cylinder.	Para. 3.1
Dimensions not as specified.	Para. 3.1.1
Construction not as specified.	Para. 3.2
Cleaning and drying not as specified.	Para. 3.3.2 and 3.3.3
Cylinders not purged with nitrogen.	Para. 3.3.3
Closure not as specified.	Para. 3.4
Tare weight not as specified.	Para. 3.5
Test date more than 1 year prior to the date of acceptance by the Government.	Para. 3.6
Markings not as specified.	Para. 3.7
Treatment and painting not as specified.	Para. 3.8
Workmanship not as specified.	Para. 3.9

4.2.2 Tests. All tests shall be completed within 1 year from date of acceptance by the Government.

4.2.2.1 Lot. A lot shall consist of not more than 200 cylinders produced from the same heat of steel, have the same treatment cycle, and be offered for delivery at the same time.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These cylinders are intended for use in the shipment and storage of chlorine. These cylinders may be adapted for shipment and storage of methyl bromide.

6.1.1 Military-unique rationale. These cylinders may be subjected to harsh environments, including temperature extremes and salt water corrosion. For this reason, they require surface preparation and painting beyond that normally provided for commercial cylinders. Additionally, the valves, that are included as an integral part of the cylinders, are constructed to standards that exceed those manufactured for commercial customers.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type of cylinder required (see 1.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2 and 2.3).
d. When a plug or valve is required (see 3.4).
e. When decalcomanias are required (see 3.7.2).
f. Blasting specification, if appropriate (see 3.8.1.1).
g. Packaging requirements (see 5.1).

6.3 Loose particulate matter in stored cylinders. Cylinders that have been cleaned, dried, valved or plugged, and subjected to storage and handling within delivery time limits, will be acceptable for chlorine service if accumulated loose ferro-related particulate matter resulting from handling does not exceed 1 gram per 1,500 cubic inches of cylinder capacity.

6.4 Part or identification number (PIN). The PIN to be used for cylinders acquired to this specification is created as follows:

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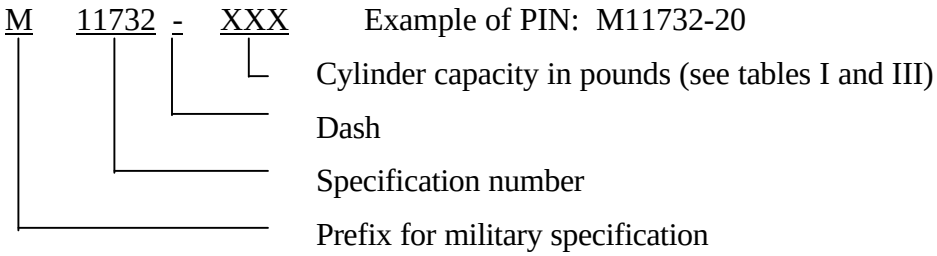

6.5 Cross-reference. The differences in cataloging data for chlorine cylinders between this revision and MIL-C-11732/1 are listed by capacity as follows:

TABLE III. Cross-reference.

Nominal gas capacity (pounds)	MIL-C-11732/1	MIL-DTL-11732E
20	C-11732/T-1	M11732E-20
30	C-11732/T-2	M11732E-30
100	C-11732/T-3	M11732E-100
150	C-11732/T-4	M11732E-150

6.6 Serial numbering. Cylinder serial numbers will be prefixed by a two or three digit symbol designating the procuring agency and will have a two-letter suffix designating the supplier furnished by the contracting officer. The serial numbers will be assigned by the supplier to the individual cylinders. These numbers may be concurrent with the supplier's regular production numbers or of a series established specifically for customer's cylinders. However, all cylinders on a given contract will be numbered consecutively and controls will be exercised to preclude duplication with future deliveries to the Government.

6.7 Subject term (key word) listings.

Cap
Flange
Methyl bromide

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Navy - SH
Air Force - 68
Army - EA

Preparing activity:
DLA - GS

Reviewers:
Navy - MC, YD1

(Project 8120-1041)

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